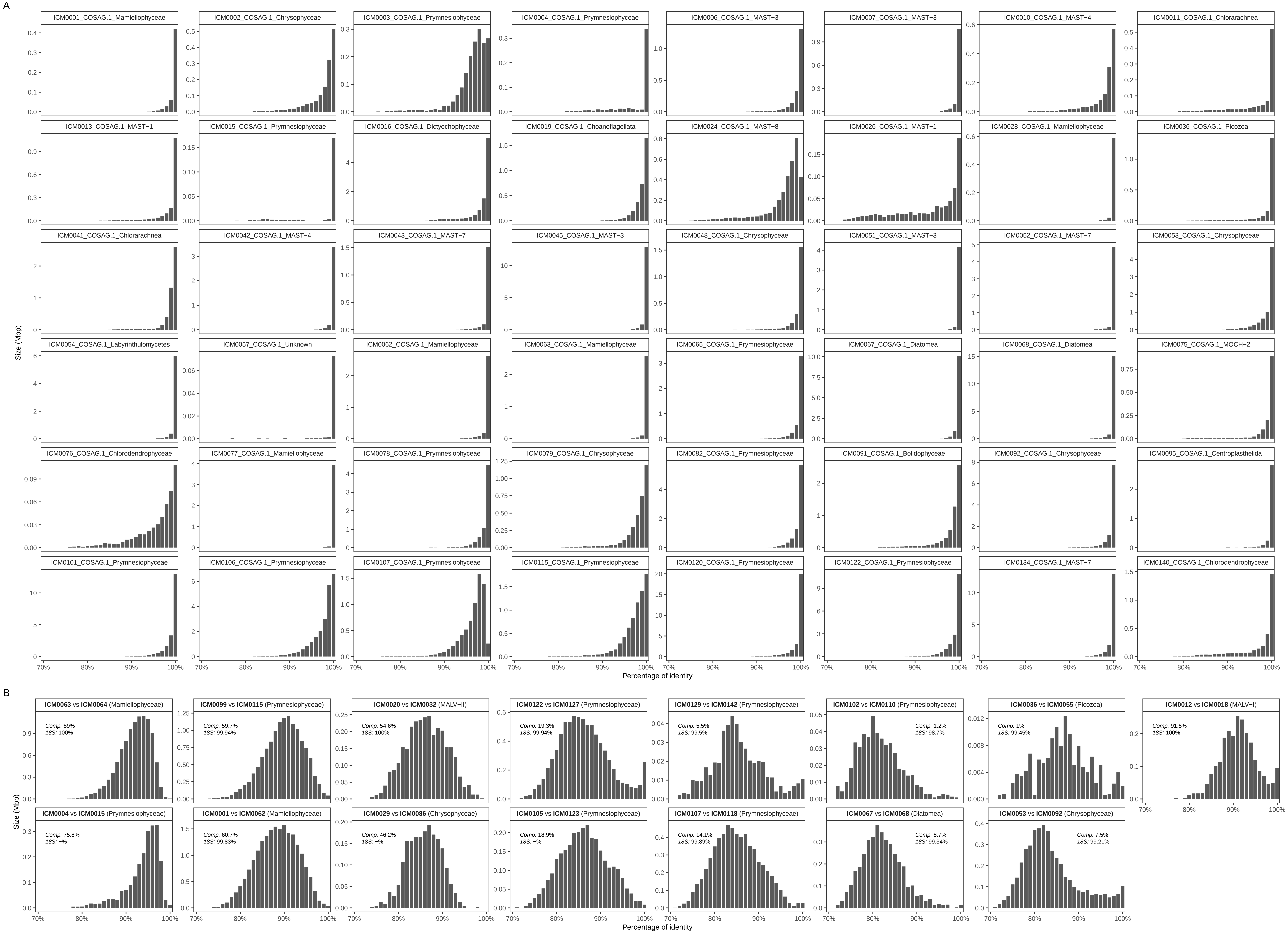

Fig. S1 | Genome similarity among SAGs within each co−assembly (A) and between closely distinct genomes (B). Pair−wise average nucleotide identity (ANI) comparisons were calculated by fragmenting the subject genome into 1 kb segments and using BLAST hits with a query >=0.5 kb. The similarity of the queried genome size was distributed in 1% bins histograms. Panels from (b), comparing different genomes, also display the sequence similarity of the 18S rDNA genes (the first 7 cases are those matching the same ASV) and the fraction of the genome that has been compared.

Fig. S2 | Full multigene phylogenetic tree of the 147 final genomes and 304 reference genomes. The tree was estimated in IQ−TREE with model LG+C60+F+G+PMSF, using a supermatrix of 240 conserved marker proteins constructed with PhyloFisher. SAGs and reference genomes are coloured by supergroup. Taxonomic groups containing SAGs are indicated. Bootstrap support is shown for all clades, with nodes highlighted in black when support is 100% and in grey when only slightly lower.

Fig. S3 | Biogeography and habitat coverage of ASVs assigned to the final genomes. Distribution of the relative abundance per sample of ASVs corresponding to the final genomes across three large−scale metabarcoding datasets: BBMO (32 samples), OSD (319 samples), and EukBank (8435 samples). The EukBank dataset is further subdivided into five major habitat categories: marine water (6545 samples), marine sediment (634 samples), freshwater (237 samples), freshwater sediment (153 samples) and soils (866 samples).

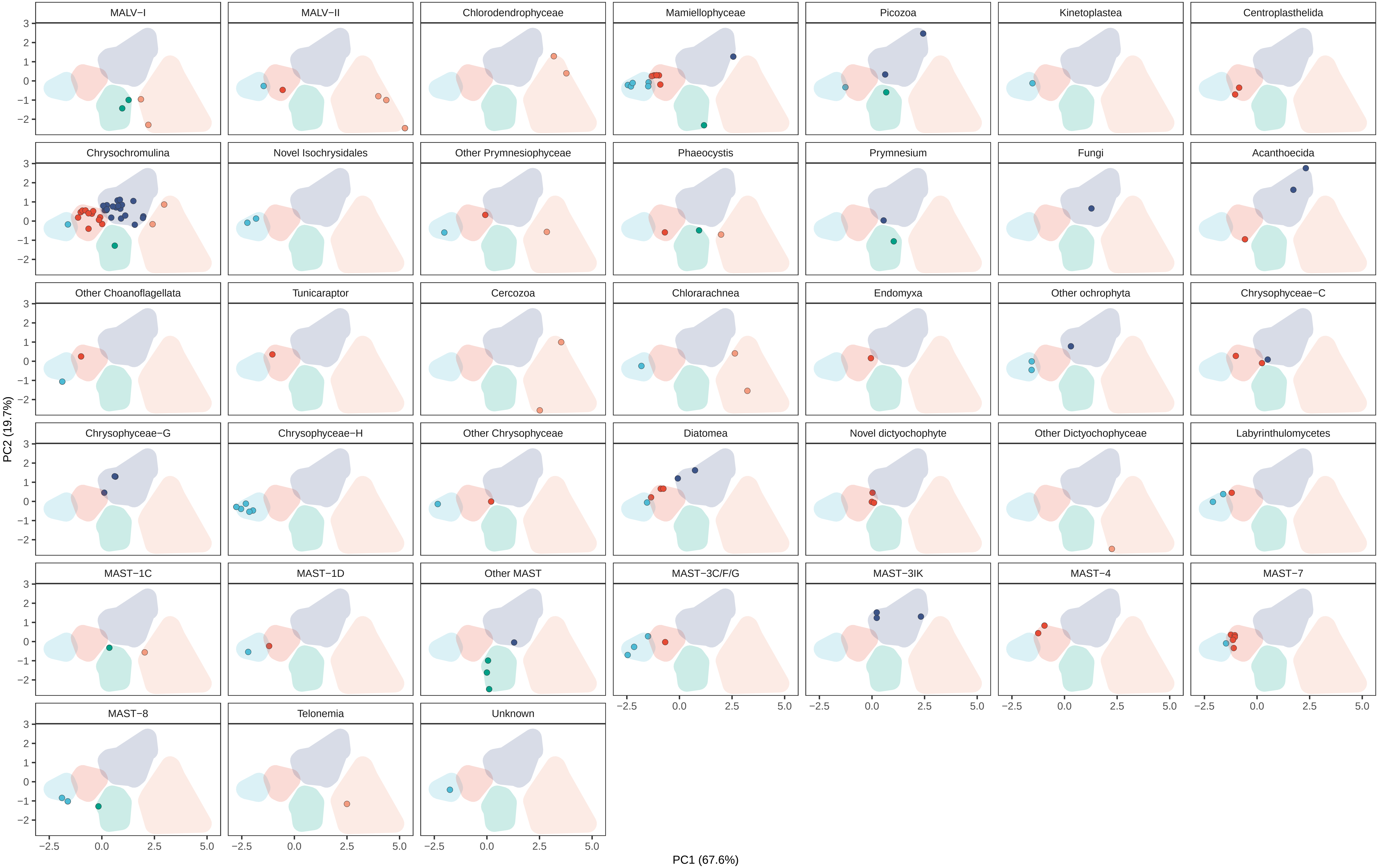

Fig. S4 | Placement of SAGs within taxonomic groups in the PCA plot shown in Fig. 4A. Separate plots are presented for each distinct taxonomic group. Shaded areas indicate k−means clusters.

Fig. S5 | Pfam domains enriched in SAGs as compared as the EukProt database of protist proteins. They are separated by being enriched only in HP, only in PP, or in the complete pool. Color intensity indicate the percentage of SAGs with the domain enriched in each subgroup.

Fig. S6 | Phylogenetic tree to place SCG rhodopsin genes. Rhodopsin genes from SAGs (158 genes; bold in the tree) were retrieved with the Pfam domain PF01036 from the gene annotations, and combined with the sequences (and group delineation) presented in Galindo et al. 2025 and with 52 MarFERReT sequences that had < 97% identity to PNAS references for a phylogenetic tree. The phylogenetic group of each sequence is indicated after the name.

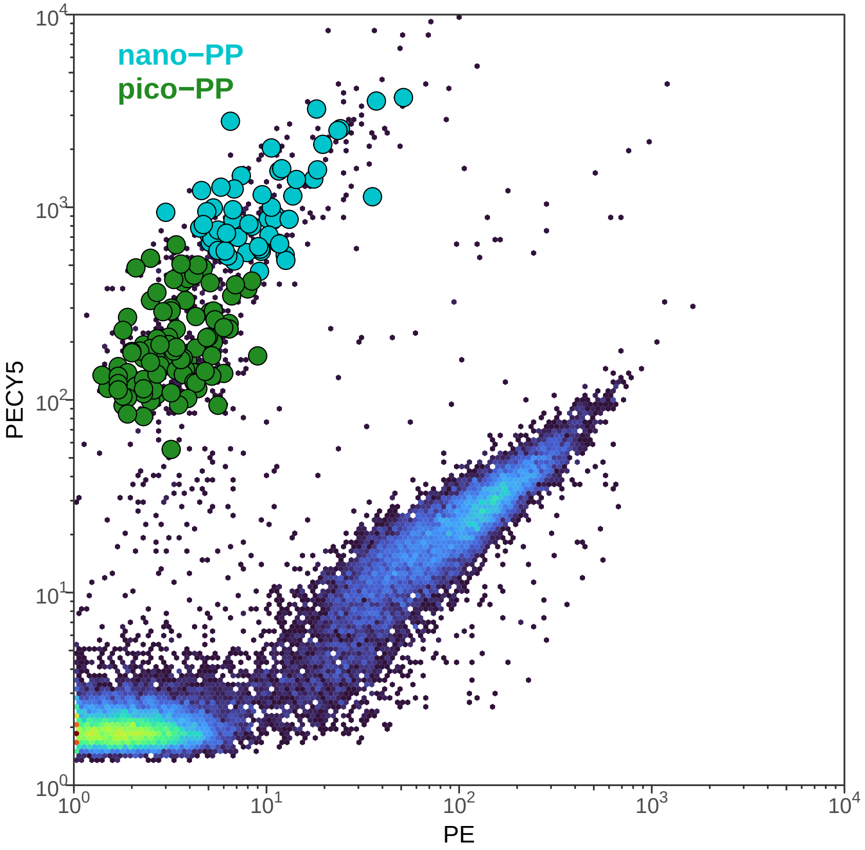

Fig. S7 | Flow cytogram of the unstained sample collected in September 2020. Large dots indicate sorted PP cells, which were subsequently classified as pico−PP and nano−PP based on flow cytometric indices.